Microeconomics 2
The Consumer Problem
Instructor Notes

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Condensed live route

The first eleven slides are unchanged. Next come exactly one Lagrangian-intuition slide, one saddle-definition slide, the forward saddle theorem with four proof obligations and its proof, then the converse saddle theorem with a four-part plan and constructive proof. A single KKT recap follows. Work through Cobb–Douglas, then Walras, substitutes, corner solutions in the linear-plus-log model and complements. Brief demand-convexity and uniqueness propositions precede indirect utility. Compute values, discuss their properties, state/prove the general envelope theorem, differentiate in wealth and prices, then prove and apply Roy. Only five reserve slides introduce expenditure minimisation.

Domain and theorem qualifications

The full Lagrangian is $\mathcal{L} = u + \lambda(m - p \cdot x) + \mu \cdot x$. In the saddle block its quantity domain is an open convex U where u is defined; the feasible set is $U \cap \{x \geq 0 : p \cdot x \leq m\}$. The forward implication needs neither smoothness nor concavity. The converse uses differentiability and concavity on U, positive prices and wealth. At a nonzero optimum construct multipliers from consumed-good returns; verify their signs, stationarity, all slackness products and both global saddle inequalities. Treat a zero optimum separately. Log Cobb–Douglas is used only after the raw continuous product index excludes axes, so U can then be the positive orthant. Never assume an undefined negative-quantity extension.

Envelope and Roy

The retained general envelope result uses C1 data and a C1 branch of optimal decisions and KKT multipliers on an open parameter neighbourhood. All choice and multiplier derivative terms are shown before stationarity, feasibility and slackness eliminate them. Roy has a separate fixed-bundle proof requiring continuous LNS utility, differentiable indirect value and positive wealth derivative, without differentiable direct utility or demand. This matters for complements and regime boundaries. The quasilinear example is a+4 log(1+q), not the earlier square-root model. Homework 1 is due 14 November 2026.

Slide-by-slide notes

1. Microeconomics: Lecture 2 — The consumer problem

Briefly identify this as the next block, without repeating the entire course administration. Homework 1 is now issued after lecture block 3, not after this block.

2. From preferences to an actual choice

Three visible bullet points only. Follow the revised order: budgets and existence, saddle/KKT and both proofs, then structural demand tools, examples, envelopes and Roy. No board-stage directions are shown to students.

3. The picture we already know

Show three utility levels (2, 4, 6) on the second reveal, including an unattainable higher curve. The third reveal adds the optimum. The budget remains p=(1,1), m=8.

4. The consumer problem

Show the maximisation objective first, then the budget inequality and all quantity nonnegativity bounds underneath it. The parameter and feasible-set definitions follow.

5. What did we assume when we wrote that?

Use the spaced bullet list; the subscription/quantity-discount/indivisibility question is a final separate reveal.

6. Currency invariance of the budget

The budget currency-invariance claim appears first, then its proof. This is not yet a statement about utility or a unique demand function.

7. The budget is nonempty and compact

The compactness claim appears before the proof. Reveal nonemptiness, closedness and boundedness separately. Strictly positive prices supply finite coordinate bounds.

8. Does an optimum exist?

No concavity or monotonicity is needed for this existence statement. Keep existence, uniqueness and computation separate throughout. The term compact means closed and bounded here because the space is finite-dimensional.

9. Which assumption is doing the work?

For the last question use u=x1+x2 at p=(1,1), any m>0: every budget-line bundle is optimal. The zero-price example is a failure example, not an assertion that all zero-price problems lack solutions.

10. Marshallian demand is initially a correspondence

Uppercase D is the correspondence and lowercase x the unique vector. This avoids using x(p,m) sometimes as a set and sometimes as a vector without warning. Marshallian demand is also called uncompensated demand.

11. Proof: demand is homogeneous of degree zero

The result is equality of correspondences, so no uniqueness assumption is necessary. A structural demand model should respect this restriction when all prices and nominal wealth are rescaled together.

12. Why introduce a Lagrangian?

Use the FULL Lagrangian from the source, with a budget multiplier and one multiplier per quantity bound. For the two saddle theorems let the decision variable range over an open convex domain U where u is defined; feasibility additionally imposes x>=0 and p*x<=m. The theorems compare feasible bundles in U. Never evaluate a log or a fractional power at undefined negative quantities.

13. What is a saddle point?

The inequalities are global on U in quantities and on the nonnegative multiplier orthant. Maximise in x at fixed multipliers, minimise in multipliers at fixed x. No max–min interchange is being assumed.

14. Theorem 1: a saddle gives a consumer optimum

This is the forward saddle theorem, not the KKT necessity theorem. The four proof obligations are primal feasibility, all complementary-slackness products, equality of saddle and utility values, and dominance over every feasible bundle in the utility domain. No concavity or differentiability is needed for this direction.

15. Proof 1: feasibility

If either a budget slack or quantity slack were negative, increasing its nonnegative multiplier arbitrarily would contradict minimisation at the saddle. Both feasibility constraints must be derived because x is varied in U rather than restricted to the orthant in this full-L formulation.

16. Proof 1: slackness and equality of values

At a feasible x, every slack is nonnegative. A positive multiplier on a strictly positive slack cannot minimise L: lowering it to zero strictly lowers the value. This proves each product vanishes and hence L(x*)=u(x*).

17. Proof 1: compare with any affordable bundle

The three-inequality chain completes the four-part proof. The first comparison uses feasible y and nonnegative multipliers, the second the global saddle, the equality complementary slackness.

18. Theorem 2: an optimum yields a saddle

This is the converse saddle theorem. Assume a differentiable concave u on an open convex ambient domain U. A consumer optimum is taken over the feasible bundles in U. The proof constructs nonnegative multipliers, verifies stationarity and slackness, then uses concavity for global maximisation in x and feasibility for minimisation over multipliers.

19. Proof 2: construct the multipliers

The construction first assumes a nonzero optimum and selects one positive quantity. Define lambda by its marginal return and mu by zero on consumed goods or lambda*p_i-u_i on zero goods. Do not assume the desired signs: those are proved next. The separate zero-bundle case is retained.

20. Proof 2: a common nonnegative spending return

A negative marginal return at a consumed quantity permits a utility-improving decrease. Different returns at two consumed goods permit an improving budget-neutral transfer. These arguments use differentiability and local feasible directions, not concavity.

21. Proof 2: why a negative bound multiplier fails

This is plain algebra and division by $p_i > 0$. In the source p.164 the left normalised derivative is mistyped: the consumed-good term must be u_j/p_j, not u_i/p_j. The same page later omits the price denominator when writing lambda for a consumed good. Both formulas here are corrected.

22. Proof 2: the profitable spending transfer

The spending transfer is one-sided because the unconsumed good can be increased and the positive holding of j can be reduced. Positive directional derivative implies a strict improvement for all sufficiently small positive epsilon by differentiability; the finite utility change is not asserted to equal its linear approximation. This proves only $\mu_i \geq 0$, not strict positivity, including at a zero quantity.

23. Proof 2: stationarity and complementary slackness

This is the missing end of the source p.164 proof, completed on pp.165 and 169. Under budget slack a consumed coordinate can move in both directions, so its derivative is zero and lambda=0. Bound products are zero by construction, not by assuming the conclusion. The source p.165 has a reversed statement about which quantities make mu zero; the correct rule is positive quantity implies zero multiplier.

24. Proof 2: maximise in quantities

For fixed multipliers, f=L is concave on U and its constructed gradient is zero. The supporting-plane inequality proves a GLOBAL maximum on U. This does not assert uniqueness or a flat function near x*. It is exactly the source p.164 reasoning, with the domain made explicit.

25. Proof 2: if the optimal bundle is zero

This gives the separate source p.162 footnote in the main proof. Small positive coordinate increases are affordable since m>0. Their derivatives cannot be positive at a local optimum. Zero budget multiplier and minus the utility gradient supply all KKT conditions.

26. Proof 2: minimise in multipliers and finish

At x* feasible slacks are nonnegative; their products with the constructed multipliers vanish. Hence L at the constructed multipliers equals u(x*), and all nonnegative multipliers give at least this value. This is the other saddle inequality. Combine it with the quantity-maximisation result. Both nonzero and zero optimum cases are covered.

27. The four Kuhn–Tucker groups

The four KKT groups summarise the construction rather than introduce another pair of theorems and proofs. The sign convention is maximisation with constraints written nonnegative. Without concavity a stationary KKT candidate need not be globally optimal.

28. Example 1: Cobb–Douglas

The original product index is continuous on the compact nonnegative budget. A boundary bundle gives zero while a positive affordable bundle gives positive utility. After excluding all axes, restrict the numerical problem to the open domain U=R++ and take logs. No smooth extension of the fractional-power index across the axes is assumed.

29. Choose a convenient representation

The log transform preserves the positive-domain ranking. Do not apply the compactness theorem directly to the open positive budget; the previous raw-utility argument already supplied existence and excluded zero coordinates.

30. Solve the Cobb–Douglas conditions

These multipliers refer to the LOG representation. The first-order equations imply lambda>0, so complementary slackness gives an exhausted budget before the later general Walras proposition. The original product utility has a different multiplier but the same demanded bundle.

31. Walras' law: an optimum exhausts the budget

This generalises budget exhaustion beyond the differentiable and concave cases. Only LNS and existence of an optimum are used. The epsilon ball is restricted to the consumption domain.

32. Example 2: perfect substitutes

Ask about spending returns and multiplicity before revealing the three regimes.

33. Verify a corner with the full KKT system

All KKT groups hold, and the objective is affine/concave, so they certify global optimality. At the corner the subjective ratio 2 is not equal to the price ratio 3, and it should not be.

34. Example 3: lessons and money left over

The new model is linear plus a shifted logarithm: $u = a + 4\log(1+q)$, $a + rq \leq m$. Price a is normalised to one. The shift allows zero consumption and makes direct utility smooth there.

35. Solve first, then respect the quantity bounds

The reduced objective is strictly concave; clip its stationary point to the feasible interval. This proves uniqueness even though the original index is affine in a. Unit-price target is q=3, replacing the old square-root example with target 4.

36. Three regimes, one decision problem

At r>=4: lambda=1, mu_q=r-4. At r<4 and m<=4-r: lambda=4/(r+m), mu_a=lambda-1, mu_q=0. In the interior lambda=1 and both bound multipliers vanish. At m=4-r the formulas coincide. At r=1, m=2 gives (0,2), m=8 gives (5,3).

37. Perfect complements: solve at the kink

Seek a useful-proportions or target-cost argument first. Smooth tangency at the kink is invalid. This is a direct utility-maximisation example, not the later full Hicksian programme.

38. Proof: convex preferences give a convex demand set

This is an ordinal result using convex preferences, not concavity of the particular utility representation. It does not establish nonemptiness; the existence assumptions are maintained.

39. Strictly convex preferences: uniqueness

Strict convexity of preferences gives at most one optimum on a convex budget. Pair with the earlier existence theorem for uniqueness. Do not confuse this ordinal condition with strict convexity of u.

40. Indirect utility: the optimal value

v is a function even if Marshallian demand is a correspondence. It is ordinally scaled: changing the representation u changes v by the corresponding increasing transform.

41. Compute the value: Cobb–Douglas

Keep log and raw cardinal scales distinct. With product exponent sum one, raw indirect utility is exp(log indirect utility). The Roy ratio is invariant under the exponential transformation.

42. Compute the value: substitutes and complements

Compute the value for both substitutes and complements directly from quantities before using any envelope or Roy theorem. The max formula includes the equal-return tie.

43. Compute the value: the logarithmic model

This is now the initial computation of the value for the a+4 log(1+q) model. Its two regime boundaries agree in value and first derivatives. Roy will be applied later using this previously computed value.

44. Indirect utility: the properties to prove

The price statement is weak, not necessarily strict for a single price. LNS gives strict increase in wealth, but a derivative can still be zero at a point. Continuity follows from the maximum theorem; it is not differentiability.

45. A less obvious property: quasiconvexity

This is joint quasiconvexity in prices and wealth, distinct from quasiconcavity of direct utility in consumption. The source proof has priming typos; the same candidate bundle x must be used at both endpoint budgets.

46. A continuous value need not be differentiable

Left derivative at p1=1 is -8, right derivative is zero. v is continuous but not differentiable. This example motivates the differentiability hypothesis in Roy's identity. It is not a failure of existence.

47. A general parameterised optimisation problem

All data are C1 on an open ambient decision/parameter domain. Quantity bounds are explicit inequalities. Parameters can enter the objective and every constraint. Equality multipliers have no sign restriction.

48. The general Lagrangian

A precise sufficient version assumes a C1 selection of optimal decisions and KKT multipliers on an open parameter neighbourhood. These assumptions need verification; they are not consequences merely of value differentiability. This is a smooth-branch theorem, not a claim covering every corner or kink.

49. Envelope theorem: a constrained smooth version

The general theorem is componentwise in a vector parameter. On the right, differentiate in the parameter holding quantities and multipliers fixed, and then evaluate at the optimum.

50. Proof: differentiate along an optimal branch

Identify V with L along the whole optimal branch using feasibility and complementary slackness. Keep the choice and multiplier derivative terms when differentiating; the next slide justifies their disappearance.

51. Proof: why multiplier derivatives disappear

For a positive inequality multiplier, its slack is zero. A zero multiplier on a differentiable nonnegative branch at an interior parameter has derivative zero. Either way g_j times lambda_j prime is zero. Equality slacks and quantity-stationarity terms vanish separately.

52. Income and price derivatives of indirect utility

Hold quantities and multipliers fixed for the partial Lagrangian derivatives, then evaluate at the optimum. Prices and wealth enter only the budget constraint. The numerical lambda depends on utility scale.

53. Roy's identity: recover demand from value

The fixed-bundle proof needs continuous LNS utility, existence, differentiable v and v_m>0. It does not assume differentiability of u or demand. The smooth consumer-envelope ratio motivates it, but the separate proof covers a broader setting.

54. Roy without differentiating demand

This proof uses continuous LNS preferences, existence, a differentiable indirect value and a positive wealth derivative. It avoids assuming demand is differentiable or a priori single-valued, and does not require concave direct utility.

55. Check Roy in the Cobb–Douglas example

The log indirect value is the logarithm of the original positive product value. Roy applies to the original continuous index and the same derivative ratio is preserved by this increasing differentiable rescaling.

Use the same log utility scale throughout this calculation. This is an actual check against a separately derived demand, not only a restatement of the identity.

56. Roy example 2: substitutes and a corner

Use a strict spending-return region for the differentiable calculation. At a tie v is not differentiable in prices and Roy cannot select a unique point from the demanded segment.

57. Roy example 4: a kink in direct utility

The min index has a kink at the optimum; v=m/(p1+p2) is smooth at positive prices and wealth. The direct Roy proof applies. Distinguish smoothness of the direct index, the indirect value and the demand vector.

58. One ratio, three regimes

At the two transition boundaries, the value is C1 even though demand has a kink. The smooth-choice-branch envelope proof may fail there, but the direct Roy proof still works. Recover q from the price derivative and a by budget exhaustion with its unit price.

59. Keep the tools, and read next

The live route ends after the Roy applications. Only five reserve slides follow, introducing next lecture extquoteright s expenditure problem; there is no long leftover proof appendix.

60. Next block: two ways to ask the consumer question

This starts the optional block-3 preview from the tail of the supplied Lecture 11. It is not required before finishing the second-block core.

61. Expenditure minimisation

Capital H distinguishes a set of minimisers from a unique Hicksian vector h. A target above the attainable utility range makes the problem infeasible; do not state unconditional existence.

62. Why an unbounded feasible set can still have a minimum

The full upper contour set need not be bounded. Truncation by the cost of one feasible benchmark is enough. Continuity here refers to the selected utility representation.

63. Expenditure: a few properties for next time

Holding the utility target fixed is essential. Concavity in prices does not require concave utility or convex preferences. Continuity on the attainable target domain is discussed under continuous utility and LNS in the instructor notes; avoid unqualified endpoint or infeasible-target claims.

64. Two identities to develop next time

This is a preview, not a full Slutsky derivation. At target u(0), the expenditure is zero and v(p,0)=u(0); interpreting this endpoint extends the value function beyond the positive-wealth core. More generally the corresponding solution sets agree under the maintained assumptions.